

LEVEL 26 INTEGRATED CONDITIONAL METHODOLOGY — TATE, MAZUR, RIBET, AND KOLYVAGIN IN LEAN 4.12

DAVID FOX

ABSTRACT. The Beal equation $A^x + B^y = C^z$ with $x, y, z > 2$ is studied here only for the signature $(4, 4, 13)$. The associated Frey curve $E : y^2 = x(x - A^4)(x + B^4)$ has displayed conductor $928 = 2^5 \cdot 29$, Néron types I_0^* at 2 and I_8 at 29, invariants c_4 and Δ , and valuations $v_2(\Delta) = 6$, $v_{29}(\Delta) = 8$. This draft records the Lean 4.12 structural verification of four honest formal surfaces and their conditional integration. Mazur: $|\mathrm{SL}_2(\mathbb{F}_{13})| = 2184$, $48 < 2184$, $13 = 2^2 + 3^2$, $288/48 = 6$, cusp Finset card 2, genus 0 (so $X_0(13)(\mathbb{Q})$ is infinite). Ribet: $928/29 = 32$, matching Finset $\emptyset$ of card 0, displayed $\dim S_2(\Gamma_0(32)) = 1$ (LMFDB 32a1), $\dim S_2(\Gamma_0(16)) = 0$, Sturm 48 and 8. Kolyvagin: rank-0 display on Cremona 26a1/26b1 with $|\mathrm{Sel}_2| = 1$, $3 \cdot 7 = 21$, $L/\Omega = 1/3$ and $1/7$, and $\mathrm{Nonempty}(\mathrm{Fintype} \alpha) \rightarrow \mathrm{Finite} \alpha$. The continued Tate boundary proves the discriminant and valuation numerics, records I_{26k} for $k = v_{29}(B + 3)$ only as the output of actual Tate-algorithm evidence, and orders Néron $32/928$, cyclotomic $q^{1/13}$ inertia, and mod-13 unramifiedness strictly forward. The integrated bridge derives proposition-valued level-32 evidence from exactly three explicit Mazur, Tate, and Ribet assumptions; it does not construct an unconditional Mathlib newform. Theorems use explicit types and `decide`, not `native_decide`. Software version DOI 10.5281/zenodo.22910461; concept DOI 10.5281/zenodo.22379293.

CONTENTS

1. Introduction	1
2. Tate — Néron conductor 928	2
2.1. Continued Tate-closure boundary	3
3. Mazur — no rational 13-isogeny	3
4. Ribet — $928 \rightarrow 32$ level lowering	4
5. Kolyvagin — MW rank-0 display on 26a1, 26b1	4
6. Lean formalization — explicit types	5
7. Computational verification	6
8. Conclusion	6
References	7

1. INTRODUCTION

The Beal conjecture asserts that if $A^x + B^y = C^z$ with integers $A, B, C \neq 0$ and exponents $x, y, z > 2$, then A, B, C share a prime factor [9]. This draft does not prove the conjecture. It records a *structural verification* of four honest Lean 4.12 displays used in the level-26 attack on the signature $(4, 4, 13)$,

$$A^4 + B^4 = C^{13}.$$

Date: 22 September 2026.

2020 Mathematics Subject Classification. 11D41, 11G05, 11F80, 68V20.

Key words and phrases. Beal conjecture, Frey curve, Tate algorithm, Mazur, Ribet, Kolyvagin, $X_0(13)$, $X_0(26)$, Lean 4, Mathlib, Mathematics of Computation.

The associated Frey curve is the displayed Weierstrass model

$$y^2 = x(x - A^4)(x + B^4) \quad \text{i.e.} \quad [0, B^4 - A^4, 0, -A^4 B^4, 0]. \quad (1)$$

The short model $[0, -(A^4 + B^4), 0, 0, 0]$ is a different curve and is not used as E .

The strategy is classical [6, 12, 11, 10]: compute the conductor of (1) by Tate’s algorithm [13], exclude a rational 13-isogeny by Mazur’s theorem on $X_0(13)$, lower the residual level $928 \rightarrow 32$ in the style of Ribet, and compare the resulting newform space with the Mordell–Weil data of $J_0(26)$ via the Cremona labels 26a1 and 26b1 [1, 8]. Each of those four names is inhabited in Lean only as a *display*: numerals, Finset cardinalities, and a typeclass upgrade. Parent files keep the corresponding algebraic statements as uninhabited proposition boundaries. The integrated release does not erase those boundaries. Its bridge combines proved Frey numerics with three named assumptions: absence of a rational 13-isogeny, identification of the conductor at 29 with the Néron conductor, and Ribet lowering from 928 to 32. The conclusion is a proposition-valued evidence record, not a modular-form object.

The formal pin is Lean v4.12.0, mathlib commit 809c3fb3b5c8f5d7dace56e200b426187516535a with inputRev v4.12.0 (not @ main), and the beal-conjecture Lake manifest prefix 83739542. The command lake update was not run. The vendor tree Level26/BealLevel26Foundations at db7a556/fea0c393 has difference 0 against b780c9c/1a00fbd. The integrated software version DOI 10.5281/zenodo.22910461 records tag v30.1.1-tate-closure-continued-43735b3; the concept DOI 10.5281/zenodo.22379293 stays, related as isVersionOf [3, 2]. The preceding software version is 10.5281/zenodo.22909486.

This is a Mathematics of Computation manuscript draft, not an arXiv preprint. There is no arXiv identifier and no arXiv badge.

The preceding four display files live in Level26/HonestB0Search/ on beal-conjecture main at ea261b9 (badge parent ff546da, merge af520d3) and on foundations phase-darmon-merel-4413 at 2b60ccd. At this writing the HonestB0Search slices are near identical; beal-conjecture is the superset, carrying the historical BealMatveevBeal target in addition to the four finals. The continued companion release is commit bb4980f, annotated tag v30.1.1-tate-closure-continued-43735b3, with exact merge base 43735b3 [5].

2. TATE — NÉRON CONDUCTOR 928

The file Tate_Frey_Conductor_29_Neron_final.lean (11c0ad7; foundations incremental 35488239286, 137s cache hit) packages the displayed conductor numeral

$$N_E^{\text{disp}} = (2 : \mathbb{N})^5 \cdot 29 = 928.$$

The named theorems displayed_Neron_conductor and final_N_E_eq_928 are that equality, proved by decide. The split displayed_928_div_29 is $928/29 = 32$.

Display 2.1 (Displayed Néron conductor). In Lean 4.12, with open Nat Finset Classical,

$$(2 : \mathbb{N})^5 \cdot 29 = 928, \quad 928/29 = 32, \quad 32 \cdot 29 = 928, \quad -29 \mid 32.$$

These are numerals. Mathlib 4.12 has no KodairaType and no NeronModel. The name $N_E = 928$ is not the conductor of a Néron model.

The displayed Weierstrass invariants of (1) are

$$c_4 = 16(A^8 + B^8 + A^4 B^4), \quad \Delta = 16A^8 B^8 (A^4 + B^4)^2.$$

On the inhabited slice one has $v_2(\Delta(1, 1)) = 6$ (type I_0^* valuation) and $v_{29}(\Delta(29, 1)) = 8$ (type I_8), together with $29 \nmid c_4(29, 1)$. The residual-level Finset $\{32, 928\}$ has card 2; that card is a Finset numeral, not two Néron models. The parent names Tate_algorithm_at_29, Frey_Neron_conductor, and Frey_conductor_29_is_Neron are *theorems in this namespace*: each re-exports the inhabited

conjunction `Tate_algorithm_at_2_and_29`. On the parent file `Tate_Frey_Conductor_29.lean` those three names stay `def Prop`.

Honesty note 2.2. The phrase “final $N_E = 928$ ” means the displayed identity $2^5 \cdot 29 = 928$. It is not Tate’s algorithm as a function on integral models. SAGE `sagemath/tate_nero_29.sage` uses the displayed Frey (1), not the short model.

Axioms on the packaged Tate final are `[propext]` for the pure numerals and `[propext, Classical.choice, Quot.sound]` for the conjunction that mentions valuations already inhabited upstream. There is no `Lean.ofReduceBool`, no `sorryAx`, and no new axiom.

2.1. Continued Tate-closure boundary. The companion module `TateCurve.Neron32_928InertiaQ13` at commit `bb4980fb0dba2c2d51c8d928935ffa647d786754` does not reinterpret the preceding numerals as geometry. It records the required local argument in the following forward order:

full $j(q)$ inversion $\longrightarrow E_q/\mathbb{Q}_{29} \longrightarrow j(E_q) = j(E_{A,B}) \longrightarrow I_{26k} \longrightarrow N_{\text{Néron}} \in \{32, 928\} \longrightarrow q^{1/13}$ inertia $\longrightarrow \bar{\rho}_{E,13}|_{I_{29}}$ where $k = v_{29}(B + 3)$ and the curve-attached parameter is required to satisfy

$$v_{29}(q) = v_{29}(\Delta) = 26v_{29}(B + 3).$$

The exact Kodaira equality is a projection from a `TateAlgorithmAt29` witness. The 32/928 equalities are projections from constructed Néron-conductor data. The cyclotomic formula is a field of the $q^{1/13}$ inertia evidence, not a consequence manufactured from divisibility alone.

The numerical surface

$$13 \mid v_{29}(\Delta), \quad 928/29 = 32, \quad 32 \cdot 29 = 928, \quad 928 \neq 32, \quad 29 \nmid 32$$

is proved independently. The theorem `tate_mod13_unramifiedness_forward` preserves that surface while projecting unramifiedness from the explicit `TateNeron32_928InertiaQ13` boundary. Thus no numerical equality is used backwards to claim Tate uniformization, Tate’s algorithm, a Néron model, a residual representation, or the cyclotomic inertia formula.

3. MAZUR — NO RATIONAL 13-ISOGENY

The file `Mazur_X0_13_No_Isogeny_final.lean` (2f4e16e; incremental 35488699511, 252s) displays the Mazur constants for $X_0(13)$ [10].

Display 3.1 (Mazur numerals).

$$\begin{array}{ll} |\text{SL}_2(\mathbb{F}_{13})| = 13 \cdot 12 \cdot 14 = 2184 & (\text{SL2_F13_card}), \\ 48 < 2184 & (\text{forty_eight_lt_2184}), \\ 13 = 2^2 + 3^2 & (\text{thirteen_eq_two_squares}), \\ 288/48 = 6 & (\text{two_eighty_eight_div_forty_eight}), \\ (\{0, 1\} : \text{Finset } \mathbb{N}).\text{card} = 2 & (\text{displayed_cusps_card}). \end{array}$$

The first four identities are axiom-free `decide` theorems. The cusp-card theorem uses `[propext, Quot.sound]`.

The parent names `X0_13_Q_infinite`, `frey_no_rational_13_isogeny`, `Serre_non_Borel_mod13`, and `mazur_no_Frey_13_isogeny` are theorems in this namespace. Each re-exports the inhabited conjunction of Display 3.1 with the genus `Nat 0`. On the parent file `Mazur_X0_13_No_Isogeny.lean` those four names stay `def Prop`.

Honesty note 3.2 (Genus 0 versus two cusps). $X_0(13)$ has genus 0, so $X_0(13)(\mathbb{Q})$ is infinite over $\mathbb{Q}$. The pasted equality $\{2 \text{ cusps}\} = X_0(13)(\mathbb{Q})$ is literature-false and is *not* a theorem. The honest Lean display is finite card 2 versus the genus-0 note `X0_13_genus_nat ≠ X0_13_cusp_count`. The theorem named `X0_13_Q_infinite` packages that conjunction; it is not a Mathlib proof that the rational points are infinite.

Geometrically, a genus-0 curve has trivial Jacobian, so $J_0(13)$ is a point (rank 0, finite). That remark is not claimed as a Lean Mordell–Weil theorem in the display file. Mazur’s theorem that the only rational 13-isogenies of elliptic curves over $\mathbb{Q}$ arise from the cusps remains `def Prop` on the parent.

Axioms: numerals `[propext]` or axiom-free; package `[propext, Quot.sound]`; LLL no-go persistence `[propext, Classical.choice, Quot.sound]`. No `native_decide`, no `sorryAx`, no new axiom.

4. RIBET — $928 \rightarrow 32$ LEVEL LOWERING

The file `Ribet_Level_Lowering_29_to_32_final.lean` (5631fd1; incremental 35489274297, 282s) displays the residual-level arithmetic of Ribet [11] without inhabiting abstract level lowering.

Display 4.1 (Ribet numerals).

$$\begin{aligned} 928/29 &= 32, & 32 \cdot 29 &= 928, & \neg 29 \mid 32, \\ 2^4 &= 16, & 16 &\mid 32, \\ \dim S_2(\Gamma_0(32))^{\text{disp}} &= 1, & \dim S_2(\Gamma_0(16))^{\text{disp}} &= 0, \\ 32 \cdot \frac{3}{2} &= 48, & 2 \cdot 48/12 &= 8, \\ (\emptyset : \text{Finset } \mathbb{N}).\text{card} &= 0. \end{aligned}$$

The displayed full/new dimensions and the Sturm pair are the LMFDB pin `full=1, new=1, level16=0, sturm=8` for the unique newform 32a1 [8].

The parent names `Ribet_928_to_32, no_newforms_at_32_mod13`, and `explicit_a29_mod13` are theorems in this namespace. Each re-exports `no_newforms_at_32`: the numerals of Display 4.1 together with `displayed_frey_matching_newforms.card = 0`. On the parent file `Ribet_Level_Lowering_29_to_32` those three names stay `def Prop`. The theorem named `explicit_a29_mod13` does not compute a_{29} .

Honesty note 4.2 (Newspace dimension 1, not 0). LMFDB records $\dim S_2(\Gamma_0(32))^{\text{new}} = 1$, with unique newform 32a1. The pasted Sage line `CuspForms(32,2).new_subspace().dimension()=0` is *not* a theorem. The honest Lean display is the empty matching Finset (card 0) versus displayed dimension 1. Witnesses: `sagemath/ribet_no_newforms_32.sage` and `certs/ribet_no_newforms_32.json` pin `full=1, new=1, level16=0, sturm=8`.

Axioms: display `[propext, Quot.sound]`; LLL no-go `[propext, Classical.choice, Quot.sound]`. No `Lean.ofReduceBool`, no `sorryAx`, no new axiom.

5. KOLYVAGIN — MW RANK-0 DISPLAY ON 26A1, 26B1

The file `Kolyvagin_MW_Rank0_26a1_26b1_final.lean` (fc8c7a1 / phase merge bc7e93b; incremental 35489926310, 262s) displays the Kolyvagin / two-descent numerals for Cremona 26a1 (LMFDB 26.a2) and 26b1 (LMFDB 26.b2) [7, 1, 8].

Display 5.1 (Kolyvagin numerals and typeclass upgrade).

$$\begin{aligned} |\text{Sel}_2|^{\text{disp}} &= 1 & (\text{Sel}_2_card_one, \text{displayed_Sel}_2_card_26a1/b1), \\ 3 \cdot 7 &= 21 & (\text{three_mul_seven}), \\ L/\Omega &= 1/3 \text{ on } 26a1, \quad 1/7 \text{ on } 26b1 & (\text{norm_num on } \mathbb{Q}), \\ \neg \text{IsRankZero} &\text{ on both models,} \\ \text{Nonempty}(\text{Fintype } \alpha) &\rightarrow \text{Finite } \alpha & (\text{finite_iff_nonempty_fintype}), \\ |\text{Unit}| &= 1 & (\text{Subsingleton card on Unit}). \end{aligned}$$

The rationals $3/9 = 1/3$ and $7/49 = 1/7$ use `norm_num`: `decide` has no $\mathbb{Q}$ instance in Lean 4.12.

Cremona 26a1 has torsion $\mathbb{Z}/3\mathbb{Z}$ and $\Delta = -17576$; 26b1 has torsion $\mathbb{Z}/7\mathbb{Z}$ and $\Delta = -1664$. The identity `IsRankZero := Subsingleton` is therefore *false* on both Mordell–Weil groups (the theorems `not_IsRankZero_26a1_final` and `not_Subsingleton_MW_26a1_final`, and the 26b1 twins). The `Subsingleton → Fintype` path applies to `Unit`, not to $E(\mathbb{Q})$.

The parent names `MW_rank_zero_fintype`, `Kolyvagin_L_nonzero_imp_MW_rank_zero`, and `TwoDescent_implies_rank_zero` are theorems in this namespace. Module-named theorems `TwoDescent_26a1_26` (from 7c19ad0), `BSD_MordellWeil`, `J0_26_BSD_26a1_26b1`, and `X0_26_Full2Torsion` package the same numerals. On the parent files, `MW_rank_zero_fintype`, `Kato`, `BSD_MordellWeil.MW_rank_zero`, and `X0_26_Full2Torsion.J0_26_rank0` stay `def Prop`.

Honesty note 5.2 (Not $E(\mathbb{Q})$ finite). Algebraic Mordell–Weil rank 0 as `Nonempty(Fintype(E(Q)))` remains `def Prop` on the parent. The packaged theorem `MW_rank_zero_fintype` is Display 5.1, not Kato–Kolyvagin and not Sage `E.rank()`. Witness: `sagemath/kolyvagin_rank0_26.sage` with `certs/kolyvagin_rank0_26.json`.

Axioms: `[propext, Classical.choice, Quot.sound]`. No `Lean.ofReduceBool`, no `sorryAx`, no new axiom.

6. LEAN FORMALIZATION — EXPLICIT TYPES

Every theorem in the four finals carries an explicit type. Lean 4.12 rejects `theorem foo := bar`. The required opens are `open Nat Finset Classical`. Typical types are $\mathbb{N}$, `Prop`, `Finset  $\mathbb{N}$` , and `Fintype`. Proofs use `decide` (or `norm_num` on $\mathbb{Q}$), never `native_decide`, so `Lean.ofReduceBool` does not appear.

Pin. Toolchain `leanprover/lean4:v4.12.0`. Mathlib `809c3fb3b5c8f5d7dace56e200b426187516535a`, `inputRev v4.12.0`. Foundations Lake manifest prefix `c7dbd578`; conjecture prefix `83739542` (package lists differ; do not run `lake update`). Vendor `db7a556/fea0c393` has difference 0 against `b780c9c/1a00fbd` and against foundations `43735b3`.

Axioms. Allowed axiom lists are `[propext]`, `[propext, Quot.sound]`, and `[propext, Classical.choice, Quot.sound]`. There is no `sorryAx` and no new axiom. Parent files keep the algebraic names as `def Prop`. Display files contain no new `def Prop`.

Integrated conditional bridge. The companion release registers the full Tate, Mazur, Ribet, and Kolyvagin surfaces together with `Conditional32Newform.Bridge` and `RequiresTateMazurRibetFor32`. Doc. The bridge proves

$$\Delta = 16A^8B^8(A^4 + B^4)^2, \quad v_{29}(\Delta) = 26v_{29}(B + 3), \quad 13 \mid v_{29}(\Delta),$$

as well as $928/29 = 32$ and $32 \cdot 29 = 928$. Its theorem `has_newform_at_level_32_of_assumptions` takes exactly `hMazur`, `hTate`, and `hRibet`. The result is `HasNewformAtLevel32`, a proposition-valued evidence record. Wiles/BCDT modularity, the geometric mod-13 representation and absolute irreducibility, Tate/Néron and cyclotomic-inertia theory, genuine Ribet lowering, identification with 32a1, and complete Kraus elimination remain explicit.

Forward Tate closure. The subsequent module `TateCurve.Neron32_928InertiaQ13` projects the exact symbol I_{26k} from Tate-algorithm evidence, the $32/928$ split from constructed Néron data, and $v_{29}(q) = v_{29}(\Delta)$ from curve-attached uniformization data. Its final mod-13 theorem consumes the proved numerical surface but obtains unramifiedness only from the explicit Tate–Néron– $q^{1/13}$ inertia boundary. In particular, $13 \mid v_{29}(\Delta)$ is never used backwards to construct the geometric representation or cyclotomic formula.

CI, incremental only. Foundations `build.yml` is `HonestB0Search` incremental: `lake exe cache get`, then `+Tate_Frey_Conductor_29_Neron_final`, `+Mazur_X0_13_No_Isogeny_final`, `+Ribet_Level_Lowering`, `+Kolyvagin_MW_Rank0_26a1_26b1_final` (~ 40 s on a warm `.lake`), then `lake build HonestB0Search` and `lake build Level26` (1–2s each). Cache key `lake-${hashFiles(lean-toolchain, lake-manifest.json)}`.

Representative greens: 35488239286 (137s, 11c0ad7); 35488699511 (252s); 35489274297 (282s); 35489926310 (262s); phase 35490246507 (303s) and 35524223408; About chore 35552037733 (01:47:20Z–01:52:05Z, 285s, .lake restored). On those working branches `main.yml` recorded 0 runs. The phase incremental 35524223408 is the green gate. This draft does not run `lake build BealMatveevBeal` and does not run `lake update`.

The integrated release ran the larger reproducibility gate: cache restoration of 5,134 files and a combined build of 5,206 jobs, including `BealMatveevBealV25B0Search`, all four `BlockerDoc` roots, the conditional bridge, and its requirements document. The publication-surface validator and `verify-matveev-beal.sh` passed. Its axiom report is `[propext,Classical.choice,Quot.sound]` with no `sorryAx`.

Conjecture `main.yml`. On `beal-conjecture` the hours job `build+HonestB0Search+Level26` runs only on `main` (and matching merge/phase/v24–v27 branches). Greens: merge-branch 35524441913 (850e1e9); main 35532850099 (af520d3); badge 35541392250 (22:20:55Z–00:58:37Z, 2h 37m) on ff546da. Tip ea261b9 is the About chore on top of that badge.

7. COMPUTATIONAL VERIFICATION

Zenodo. The integrated software version DOI 10.5281/zenodo.22910461 records the v30.1.1 Tate-closure release [3]. The preceding software version is 10.5281/zenodo.22909486 [4]. The concept DOI 10.5281/zenodo.22379293 stays, recorded as `isVersionOf` [2]. The corresponding foundations GitHub release is at bb4980f [5]; the predecessor monorepo tag v30.0.0-level-26-structural-verification remains at af520d3.

Sage and JSON.. The Lean displays are paired with static Sage scripts and JSON pins, not with a Sage kernel in CI: `sagemath/tate_nero_29.sage`; `sagemath/ribet_no_newforms_32.sage` with `certs/ribet_no_newforms_32.json` (full=1, new=1, level16=0, sturm=8); `sagemath/kolyvagin_rank0_26.sage` with `certs/kolyvagin_rank0_26.json`. LMFDB labels used as pins are 32a1, 26a1, and 26b1 [8]. Repository surface. Folder READMEs are complete outside the vendor tree (difference 0 against 43735b3). GitHub About texts are 302 characters (foundations, homepage the concept DOI) and 308 characters (conjecture, homepage the version DOI). Topics include `lean4`, `mathlib`, `beal-conjecture`, `frey-curve`, `tate-algorithm`, `mazur`, `ribet`, `kolyvagin`, `modular-curves`, `number-theory`, `mcom` (conjecture also x0-13, x0-26). There is no arXiv badge.

Tags and working branches. Foundations tags v0.27, v0.28, v0.29, v0.30 stay live. Phase `phase-darmon-merel-4` stays 2b60ccd (v0.30-mcom-structural-verification). Working finals stay `tate-neron-final-v29` 11c0ad7, `mazur-no-isogeny-final-v29` 2f4e16e, `ribet-928-to-32-final-v29` 5631fd1, `kolyvagin-rank0-final-v29` fc8c7a1. Conjecture tag v30.0.0-level-26-structural-verification stays at af520d3; main at this draft's parent is ea261b9.

8. CONCLUSION

Level 26 structural verification records four honest formal surfaces: Tate $928 = 2^5 \cdot 29$ (I_0^* , I_8), Mazur 2184 and cusp card 2 versus genus 0, Ribet $928/29 = 32$ with matching Finset $\emptyset$ against LMFDB 32a1 of dimension 1, and Kolyvagin $|\text{Sel}_2| = 1$, $3 \cdot 7 = 21$, $L/\Omega = 1/3$ and $1/7$ on 26a1/26b1. The software versions are v30.0.0-level-26-structural-verification on `beal-conjecture` (af520d3, About tip ea261b9) and v30.1.1-tate-closure-continued-43735b3 on foundations bb4980f, with 10.5281/zenodo.22909486 as the preceding software version and v30.0.0-level-26-structural-verification as the monorepo predecessor. Phase stays 2b60ccd. Tags v0.27–v0.30 stay live. The bridge is conditional on the named Mazur, Tate, and Ribet inputs; the analytic Kolyvagin line remains separate. Parent algebraic statements remain explicit proposition boundaries.

This draft is a Mathematics of Computation submission, not an arXiv preprint. Cite the version DOI 10.5281/zenodo.22910461 and the concept DOI 10.5281/zenodo.22379293.

REFERENCES

1. John E. Cremona, *Algorithms for modular elliptic curves*, 2 ed., Cambridge University Press, 1997.
2. David Fox, *Beal level 26 foundations (concept)*, Zenodo, 2025, Concept DOI; `isVersion0f` for the v30 record.
3. ———, *Formal verification of the level 26 foundations for signature (4,4,13)*, Zenodo, 2026, Software version `v30.1.1-tate-closure-continued-43735b3`; forward Tate closure through I_{26k} , Néron 32/928, $q^{1/13}$ inertia, and mod-13 unramifiedness boundaries.
4. ———, *Formal verification of the level 26 foundations for signature (4,4,13)*, Zenodo, 2026, Software version `v30.1.0-integrated-full-43735b3`; integrated Tate–Mazur–Ribet–Kolyvagin conditional level-32 methodology.
5. ———, *beal-level-26-foundations tag v30.1.1-tate-closure-continued-43735b3*, <https://github.com/DavidFox998/beal-level-26-foundations/releases/tag/v30.1.1-tate-closure-continued-43735b3>, 2026, Commit `bb4980fb0dba2c2d51c8d928935ffa647d786754`; exact merge base `43735b3b14a9e844d4b4bdf57fa5f68d29e749ee`.
6. Gerhard Frey, *Links between stable elliptic curves and certain Diophantine equations*, Annales Universitatis Saraviensis. Series Mathematicae **1** (1986), no. 1, 1–40.
7. Victor A. Kolyvagin, *Euler systems*, The Grothendieck Festschrift, Vol. II **87** (1990), 435–483.
8. LMFDB Collaboration, *The L-functions and modular forms database*, <https://www.lmfdb.org>, 2026, Labels `32a1`, `26a1` (LMFDB 26.a2), `26b1` (LMFDB 26.b2).
9. R. Daniel Mauldin, *A generalization of Fermat’s last theorem: The Beal conjecture and prize problem*, Notices of the American Mathematical Society **44** (1997), no. 11, 1436–1437.
10. Barry Mazur, *Rational isogenies of prime degree*, Inventiones Mathematicae **44** (1978), no. 2, 129–162.
11. Kenneth A. Ribet, *On modular representations of $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ arising from modular forms*, Inventiones Mathematicae **100** (1990), no. 2, 431–476.
12. Jean-Pierre Serre, *Sur les représentations modulaires de degré 2 de $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$* , Duke Mathematical Journal **54** (1987), no. 1, 179–230.
13. John Tate, *Algorithm for determining the type of a singular fibre in an elliptic pencil*, Modular Functions of One Variable IV, Lecture Notes in Mathematics, vol. 476, Springer, Berlin, 1975, pp. 33–52.

OPERA NUMERORUM